TOSIN KOLAJO GBADEGESIN

Ph.D. in Economics • Energy, Environmental and Urban Economics • Causal Inference, Applied Econometrics and Data Science
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U.S. Work Authorized (No Sponsorship Required)

AREAS OF EXPERTISE

Energy and Environmental Economics	Urban and Spatial Economics	Labor and Welfare Economics
Causal Inference and Program Evaluation	Econometric Modeling and Forecasting	Heterogeneous Treatment Effects
Applied Machine Learning	Policy Analysis and Translation	Large-Scale Data Engineering

PROFESSIONAL SUMMARY

I am an applied microeconomist (Ph.D. in Economics) specializing in energy, environmental, and urban economics, with expertise in causal inference, applied machine learning, and large-scale quantitative policy analysis. My research focuses on how energy markets, environmental regulations, and climate change affect household welfare, economic opportunity, and public policy. At the World Bank Group, I design and deploy empirical research from econometric early warning systems for food insecurity to causal evaluations of large-scale investment policies. My dissertation, *Climate Change, Energy Demand, and Policy Impacts: Analyzing Residential and Building Energy Dynamics in the U.S.*, applies causal inference and distributional methods to energy demand, climate resilience, and policy evaluation. I have published in several peer-reviewed journals, including Applied Energy, Energy and Environment, and Defence and Peace Economics. I am equally committed to communicating research findings to technical and non-technical audiences, having presented at the World Bank Annual Meetings, the ASSA/AEA Annual Meetings, and the Germany-Africa Head of State Summit.

EDUCATION

Ph.D. in Economics, Howard University, Washington, DC *2026*

- Fields: Energy and Environmental Economics, Urban and Regional Economics; Industrial Organization, Causal Inference
- Dissertation: "Climate Change, Energy Demand, and Policy Impacts: Analyzing Residential and Building Energy Dynamics in the U.S."
- Job Market Paper (Chapter 4): "The Building Energy Performance Standard Program: Impacts on Buildings' Energy Efficiency and GHG Emissions in Washington, DC," Applied Energy. [Link], [Replication Code]
- Frederick Douglass Doctoral Scholar (2023-24) | World Bank Africa Doctoral Fellow; 1 of 32 selected from 3,000+ applicants worldwide

M.Sc. in Economic Development, University of Ibadan, Nigeria *2018*

B.Sc. in Microbiology, Obafemi Awolowo University, Nigeria *2013*

PROFESSIONAL EXPERIENCE

Economist *January 2023 – Present*

World Bank Group • Washington, DC

- Lead the design of a Private Sector Gender and Economic Inclusion Data Platform comprising statistical frameworks, automated ETL pipelines, and BI dashboards processing 10-14 TB of multi-source data to support the World Bank's global reporting and operational decision-making.
- Directed quasi-experimental evaluation of the G20 Compact with Africa (spanning 12 African economies) using Difference-in-Differences and PSM; quantified 5-10% gains in FDI and domestic financing and delivered findings at the Germany-Africa Head of State Summit, directly informing sovereign investment policy.
- Developed structural VAR and error-correction econometric models to quantify causal linkages between climate shocks, food prices, and agricultural wages; analysis operationalized by the World Bank's Early Response Financing mechanism, extending intervention lead times by up to 12 months in fragile and conflict-affected states.
- Produced applied economic research on electricity market structure and extractives sector performance; refined IFC's ex ante impact measurement framework to link power market design to investment and development outcomes across emerging markets.
- Built ensemble ML models (LightGBM, XGBoost, CatBoost, HistGradient Boosting) on 500+ GB historical lending data, improving non-performing loan detection by 15% and strengthening regulatory-grade portfolio risk monitoring.

Research Assistant | Teaching Assistant *2024 – 2025*

Howard University Graduate School • Washington, DC

- Processed 100M+ UN Comtrade trade records using Python and SQL for faculty international trade research; Teaching Assistant for Economic Statistics covering data-lab instruction and applied econometrics.

Energy Efficiency Analyst *June – December 2022*

DC Sustainable Energy Utility (DCSEU) and DC Dept. of Energy and Environment (DOEE) • Washington, DC

- Conducted regulatory benefit-cost and causal policy analysis of DC's Building Energy Performance Standards across 100+ GB of commercial building panel data; quantified 5-10% efficiency gains; translated findings into regulatory guidance and stakeholder policy briefs.

Research Analyst *September – December 2022*

Edelman Global Advisory • Washington, DC

- Delivered quantitative economic and policy analysis on trade and climate policy across six global regions; produced structured research reports informing client decisions in markets representing $500B+ in annual trade.

ADDITIONAL EXPERIENCE

Bridge Policy Research Fellow, ***Nigerian Economic Summit Group*** • Abuja, Nigeria *August 2020 – August 2021*

- Analyzed COVID-19 socioeconomic impacts and national development frameworks; co-led establishment of the UN CEO Water Mandate in Nigeria; integrated gender and digital technology themes into medium-term national development plans and Agenda 2050.

Development Finance Specialist, ***Jerusalem Institute for Policy Research | Gigawatt Global Inc.*** • Israel*January – September 2020*

- Assessed economic feasibility and ESG impacts of grid-connected solar projects in emerging markets; built financial models evaluating investment returns, risks, and capital requirements; engaged development finance institutions to mobilize project capital.

Data Analyst, ***International Institute for Local Development*** • Abuja, Nigeria *September 2019 – July 2021*

- Applied ML models (Random Forest, AdaBoost, KNN, SVM, CNN) to evaluate World Bank-funded rural access and agricultural development projects; produced analytical reports and Tableau dashboards for decision-makers.

Research Fellow, ***International Fund for Agricultural Development*** • Arusha, Tanzania *January – June 2018*

- Investigated smallholder farmers' market access barriers using survey econometrics; published findings in the Africa Journal of Sustainable Development.

Research Fellow, ***Africa Water Association and USAID West Africa*** • Nigeria *July – December 2018*

- Designed survey data collection pipelines (Kobo Collect), analyzed in SPSS, built Tableau dashboards, and presented findings at an international sustainable development conference.

TECHNICAL SKILLS

Econometrics and Causal Inference: Panel and Time-Series Models; IV and GMM; Spatial Econometrics; DiD; RDD; Synthetic Control; PSM; Quantile Regression; TMLE; Causal Forests; Double ML; CATE/CDTE and HTE Estimation

Machine Learning and Data Science: XGBoost, LightGBM, CatBoost, Random Forest, AdaBoost, SVM, CNN, Super Learner; NLP and LLMs; Feature Engineering; Large-Scale Data Processing; Automated Pipelines; A/B Testing

Programming and Tools: Python (pandas, statsmodels, scikit-learn, EconML); R; Stata; SQL; MATLAB; Power BI; Tableau; ArcGIS; SPSS; PVSYST; LaTeX; GitHub; Azure; AWS

Domain Expertise: Energy and Environmental Economics; Regulatory and Policy Economics; Labor and Urban Economics; Development Finance; Gender and Inclusion Data; ESG and Impact Measurement

RESEARCH AND PUBLICATIONS

Published

- "The Building Energy Performance Standard Program: Impacts on Buildings' Energy Efficiency and GHG Emissions in Washington, DC", *Applied Energy*. [Link], [Replication Code]
- "Spatial Determinants of Rent Burden: Evidence from the United States' Census Tracts", *Economic Development Quarterly*. **[Accepted]**
- "Defense Expenditure and National Well-Being: Evidence from the G7 Countries", *Defence and Peace Economics*, 2024, 36(7), 1030-1041. [Link]
- "Evaluating the Economic Impacts of the G20 Compact Initiative: Evidence from Causal Inference Using Advanced Machine Learning Techniques" (with N. D. Yameogo), *Journal of Sustainable Development*, 2024, 17(5), 56-76. [Link]
- "Climate Change and Energy Dynamics: Evidence from BRICS and G7 Countries", *Energy and Environment*, 2024, 0(0). [Link]
- "Climate Shocks and Their Effects on Food Security, Prices, and Agricultural Wages in Afghanistan" (with B. P. J. Andree and A. Braimoh), *Policy Research Working Paper, World Bank Group (PLANET)*, 2024. [Link]

Under Review

- "Heterogeneous Effects of Cooling Demand on Household Electricity Consumption using Causal Machine Learning Models", *Applied Energy*.
- "Impact of Climate-Driven Cooling Demand on U.S. Household Electricity Consumption and Expenditure", *Energy and Environment*.

World Bank Group Reports

- Closing the Gap: A Data Outlook on Women's Economic Opportunities, *International Finance Corporation*, 2025. [Link]
- Repurposing Power Markets: The Path to Sustainable and Affordable Energy for All, *International Finance Corporation*, 2024. [Link]
- G20 Compact with Africa Monitoring Report, *World Bank Group*, 2023. [Link]

HONORS, AWARDS, AND PROFESSIONAL ACTIVITIES

World Bank Group Africa Doctoral Fellow; 1 of 32 selected from 3,000+ global applicants	*2023*
Frederick Douglass Doctoral Scholar, Howard University Graduate School	*2023*
Blum Global Fellow in Development Finance, Jerusalem Institute for Policy Research, Israel	*2020*
Early Career Conference Grant, Association of Commonwealth Universities (ACU), London	*2019*
YALI Leadership Award, Energy and Environmental Sustainability	*2018*

Conferences: NABE Tech Economics Conference, Seattle (2025); NABE Economic Measurement Seminar, Washington DC (2025); Empirical Methods in Energy Economics, World Bank (2025); ASSA/AEA Annual Meetings (2024); Germany-Africa Head of State Summit (2023); World Bank Annual Meetings (2023); Development Studies Association, Dublin (2019).